

# CERTIFIED ROBUSTNESS FOR TEMPORAL GNNs VIA SHARED-NOISE SMOOTHING

Rohan Gupta<sup>†</sup> Jeevesh Mahajan<sup>†</sup>

<sup>†</sup>University of Wisconsin–Madison | CS 763: Trustworthy AI | {rk Gupta3, jmahajan2}@wisc.edu



## Problem & Motivation

Graph Neural Networks are the standard architecture for fraud detection on financial transaction graphs, yet structural attacks like NETTACK [1] flip predictions with a **median of just 2 edge edits**. Heuristic defenses (GNN-Guard, ProGNN) lose 15–30 pp under adaptive attack [2] and offer *no* formal guarantees.

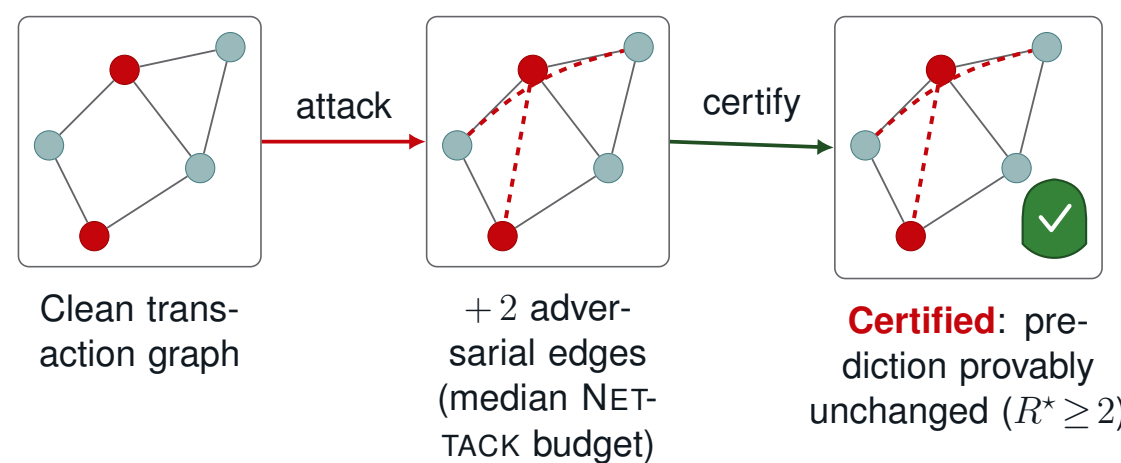


Fig. 1: **The threat, in three snapshots.** Clean graph → adversarial edits → provably robust prediction.

Real fraud graphs are also **temporally evolving and heterophilous**. Existing certified-robustness work for GNNs targets a single static snapshot under uniform Bernoulli noise — destroying the cross-class signal that fraud detection relies on, and leaving the temporal-attack surface uncertified.

**Goal.** A certified-robustness guarantee that (i) holds across a *window* of timesteps, (ii) respects heterophilous structure, and (iii) is tighter than a per-window union bound.

## The Three-Phase Project Arc

### Phase 1 – Threat surface.

Trained GCN/GAT/GraphSAGE on a 12k-node synthetic fraud graph (edge homophily 0.42). NETTACK flipped all 20 targets at median 2 edges; PGD at 5% budget collapsed illicit-F1 from 0.80 to 0.13. Uniform 20% smoothing also dropped illicit-F1 to 0.65 — noise itself was destroying the signal.

### Phase 2 – Heterophily-aware noise.

Uniform smoothing wipes out the cross-class edges fraud detection needs, so we use locality-aware noise: preserve boundary-crossing edges with lower flip rates and add more noise to same-class edges. This enables the per-snapshot Neyman–Pearson certificate.

### Phase 3 – Shared-noise temporal certificate.

Replaces per-window Bonferroni with a single softmax-sum statistic on one shared randomness source. Strictly tighter, sound, and benchmarked across four datasets.

## Background: Randomized Smoothing

We turn a vulnerable base classifier into a provably robust one by averaging its predictions over many randomly perturbed copies of the input graph.

Let  $f$  be a trained Graph Convolutional Network,  $G$  be the input graph, and  $\Pi$  be a noise distribution that independently flips each edge with some probability. A single random perturbation  $\xi \sim \Pi$  produces a noisy graph  $\Delta_\xi(G)$ . The *smoothed classifier* is the plurality vote across these noisy realizations:

$$g(v) = \arg \max_c \Pr_{\xi \sim \Pi} [f(\Delta_\xi(G))(v) = c]. \quad (1)$$

For each evaluated node  $v$ ,  $N$  smoothing samples give a Clopper–Pearson lower bound  $\underline{p}_A$  on the top-class probability at confidence  $\alpha$ . The Neyman–Pearson construction of Bojchevski & Günnemann [3] yields the *certified radius*

$$R^* = \max \{ R \in \mathbb{N} : \text{NP test rejects all } \Pi' \text{ at edit dist. } \leq R \}. \quad (2)$$

The prediction  $g(v)$  is provably unchanged under *any* adversary that flips at most  $R^*$  edges.

## Three Certificate Variants

All three variants share the same trained Graph Convolutional Network and the same smoothing hyperparameters: an existing edge is flipped with probability  $p_{\text{smooth}} = 0.10$ , a non-edge is added with probability  $p_{\text{add}} = 0.01$ , the certificate’s target failure rate is  $\alpha = 0.01$ , and the temporal window count is  $T = 3$ . They differ only in *how* the per-window certificates are composed:

|                                |                                                                                                                                                                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aggregate</b>               | A single Neyman–Pearson certificate computed once on the full static graph at the target failure rate $\alpha$ . Ignores the temporal split entirely.                                                                                                |
| <b>Per-window (Bonferroni)</b> | One independent Neyman–Pearson certificate per window, each tightened to a stricter coverage $\alpha/T$ so the union bound keeps total failure $\leq \alpha$ . Sound, but the union bound is tight only when the per-window outputs are independent. |
| <b>Shared-noise (ours)</b>     | One edge-flip mask $\xi$ is drawn and reused across all $T$ windows. Per-window softmax outputs are summed into a single score vector per node, then closed with one Clopper–Pearson bound at the full coverage $\alpha$ - no Bonferroni penalty.    |

## Shared-Noise Construction

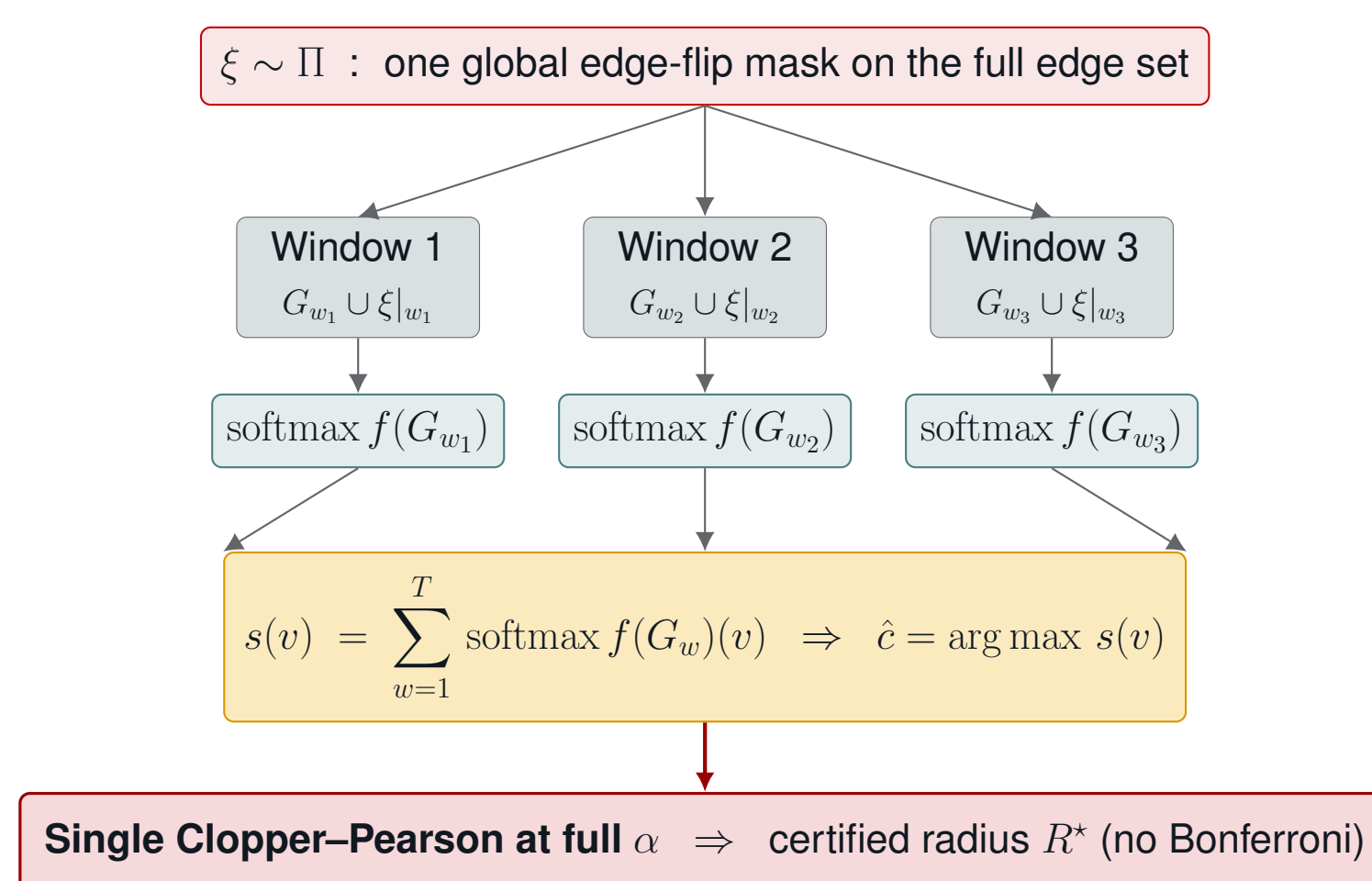


Fig. 2: **Shared-noise temporal certificate.** One global edge-flip mask  $\xi$  feeds  $T$  window-restricted forwards; per-window softmax outputs are summed before a *single* Clopper–Pearson bound at full  $\alpha$ .

**Why it is sound.** The certificate is a deterministic function of *one* randomness source  $\xi$ , so the standard Neyman–Pearson argument applies to the aggregated statistic exactly as to a single forward pass.

**Why it is tighter.** Per-window outputs from one trained model on overlapping graph regions are **positively correlated**. The Bonferroni union bound is tight only under independence; shared-noise exploits the actual correlation, *strictly* dominating Bonferroni whenever it is positive.

## Heterophily-Aware Bernoulli Noise

Uniform edge noise destroys the cross-class signal that fraud detection depends on. We replaced the uniform noise with a noise distribution whose per-edge flip probability  $p_e$  is a function of how similar the two endpoints are. **Cross-class edges** are perturbed gently to keep that signal intact, while **same-class edges** absorb more noise, giving the certificate a wider effective radius without eroding accuracy.

On the synthetic fraud graph this produced flip probabilities in  $[0.050, 0.150]$  with mean  $p_e = 0.106$  (0.077 on same-class edges, 0.128 on cross-class edges).

## Headline: Certified Accuracy at $R=2$

$R=2$  matches the median NETTACK budget measured in Phase 1, so it is the operationally relevant radius. Shared-noise **strictly beats both alternatives in every experiment**.

| Dataset                  | Hom. | Agg.     | Bonf.    | Shared                 |
|--------------------------|------|----------|----------|------------------------|
| Synth. fraud (N=1000)    | 0.44 | 0.383    | 0.133    | <b>0.500</b>           |
| Synth. fraud (N=2000)    | 0.44 | 0.561    | 0.233    | <b>0.722</b>           |
| Cora-temporal            | 0.81 | 0.238    | 0.095    | <b>0.405</b>           |
| Real Elliptic Bitcoin    | 0.95 | 0.267    | 0.383    | <b>0.683</b>           |
| $\Delta$ (Shared – Agg.) | –    | +0.117 / | +0.161 / | +0.167 / <b>+0.417</b> |

Per-window Bonferroni actually *underperforms* the static aggregate on 3/4 experiments — the union-bound penalty wipes out whatever per-window tightening the temporal split provides. **Sanity check:** re-running synthetic fraud at  $N=2000$  widens the gap from +0.117 to +0.161, ruling out a small- $N$  Clopper–Pearson artefact.

## Real Elliptic Bitcoin: Credibility Check

Elliptic is the toughest test in the matrix:  $\sim 204k$  nodes, 49 weekly timesteps, edge homophily 0.954, and *real* adversarial transaction dynamics. Test windows are  $w_1=[35, 39]$ ,  $w_2=[40, 44]$ ,  $w_3=[45, 49]$ .

| Variant             | $R=0$        | $R=1$        | $R=2$        | $\bar{R}^*$ |
|---------------------|--------------|--------------|--------------|-------------|
| Aggregate           | 0.667        | 0.600        | 0.267        | 1.18        |
| Per-window (Bonf.)  | 0.667        | 0.617        | 0.383        | 1.40        |
| <b>Shared-noise</b> | <b>0.717</b> | <b>0.700</b> | <b>0.683</b> | <b>1.90</b> |

$\bar{R}^*$ : average certified radius — the average number of edge edits each prediction is provably guaranteed to tolerate.

Per-window, shared-noise certified accuracy at  $R=2$  falls only from 0.60 on  $w_1$  to 0.25 on  $w_3$ , while Bonferroni falls below 0.10. Shared-noise **degrades most gracefully under temporal drift**.

## Conclusions & Future Work

- Sound and strictly tighter.** The shared-noise certificate keeps the standard Neyman–Pearson soundness guarantee while replacing per-window Bonferroni’s  $\alpha/T$  penalty with a single Clopper–Pearson bound at full  $\alpha$ .
- Dataset-agnostic gains.** The same construction wins across the entire homophily spectrum — synthetic fraud (0.44), Cora (0.81), Elliptic Bitcoin (0.95) — and across two orders of magnitude in graph size, with a peak improvement of **+41.7 percentage points** at  $R=2$  on the only experiment whose temporal dynamics are real.
- Future work.** Sweep  $p_{\text{smooth}}$  for full accuracy–radius Pareto curves; vary  $T$  (the Bonferroni penalty grows with  $T$ , so the gap should widen); benchmark adaptive attackers that target the realised mask  $\xi$ ; and combine the Phase 2 heterophily-aware noise with the shared-noise sampler into a single deployment recipe.

## References & Acknowledgements

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